

Investor Deep Dive

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Idea 1 Investor Deep-Dive v0

Concept: BTC Volatility Intermediation Desk — ETF / Deribit / CME / OTC volatility analytics, treasury yield/protection, and miner hedging.

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Status: Investor-grade first draft, but not final. Data evidence is screen-only unless explicitly marked otherwise. Legal/regulatory structure requires counsel before any execution or client-facing activity.

1. Executive thesis

Bitcoin has become institutionally accessible through multiple wrappers: spot ETFs, CME futures/options, crypto-native options, and OTC derivatives. These wrappers reference substantially similar economic exposure but trade through different legal, margin, collateral, custody, and client-flow regimes.

That fragmentation creates a business opportunity: build an institutional BTC risk-intermediation platform that helps treasuries, miners, funds, and allocators transform BTC volatility into yield, protection, and structured risk.

The desk should not be framed as a generic crypto-options prop fund. The higher-quality institutional wedge is:

> Board-ready BTC treasury and miner hedging programs, powered by a proprietary cross-venue BTC volatility monitor.

The analytics engine normalizes ETF options, CME futures/options, Deribit options, and OTC/RFQ quotes into BTC-equivalent terms. The client-facing business translates that intelligence into covered calls, collars, put spreads, production hedges, and eventually structured notes.

2. Why now

2.1 BTC moved into institutional wrappers

The BTC market is no longer only spot exchange flow. Institutions can now access BTC exposure through:

- US spot BTC ETFs and listed ETF options.
- CME Bitcoin futures and options.
- Crypto-native options venues such as Deribit.

- OTC bilateral derivatives and structured products.

Each wrapper attracts different users and constraints:

- ETF options: securities accounts, RIAs, family offices, public-company treasury comfort, OCC-listed options.
- CME: regulated futures accounts, FCM margin, institutional futures infrastructure.
- Deribit: deep crypto-native vol surface, broad strikes/expiries, 24/7 market, crypto collateral.
- OTC/RFQ: custom tenors, notional sizes, payoff shapes, legal/collateral terms.

2.2 BTC treasury companies and miners create natural client flow

BTC treasury companies and miners are structurally long BTC. But they also face fiat obligations: debt, preferred dividends, operating expenses, power costs, ASIC purchases, site capex, and public-market dilution pressure.

They need a framework that lets them:

- monetize volatility without selling core BTC;
- buy disaster protection without abandoning upside;
- protect capex/runway windows;
- reduce forced-equity or forced-BTC-sale risk;
- explain the policy to boards, auditors, lenders, and investors.

2.3 Volatility fragmentation creates pricing and hedging intelligence

A BTC option is not just a BTC option. The economic comparison depends on:

- ETF wrapper mechanics;
- futures basis;
- BTC/share conversion;
- contract multipliers;
- fees and expense drag;
- bid/ask and executable size;
- margin and collateral currency;
- custody/venue eligibility;
- assignment and settlement mechanics;
- regulatory/mandate restrictions.

Most simple IV comparisons ignore these. The desk's edge is to normalize them.

3. Core product menu

3.1 BTC treasury covered-call program

Client: BTC treasury companies, operating companies with BTC reserves, wealth platforms holding BTC ETFs.

Structure: Sell calls against a defined minority tranche of BTC/ETF exposure.

Purpose: Generate recurring premium income without selling spot BTC.

Variants:

- monthly OTM covered-call ladder;
- quarterly covered-call program;
- call-spread overwrite to preserve extreme upside;
- BTC-denominated premium reinvestment;
- ETF-option version for securities-account clients.

Revenue model: Structuring fee, execution commission/spread, recurring monitoring/reporting retainer, later principal risk spread if permitted.

Key control: Do not cap too much upside. Keep notional modest and policy-driven.

3.2 BTC treasury collar / downside floor

Client: BTC treasury companies, family offices, miners with BTC inventory, firms with debt/preferred obligations.

Structure: Buy puts or put spreads; finance partly/fully by selling calls or call spreads.

Purpose: Create a board-approved downside floor while preserving strategic BTC ownership.

Variants:

- 3-month zero-cost collar;
- 6-month put-spread collar;
- debt/coupon/refi-window protection;
- LTV/collateral protection around BTC-backed loans;
- drawdown-budget hedge for public-company boards.

Revenue model: Structuring fee, RFQ/execution spread, advisory retainer, risk recycling.

Key control: Define maximum BTC encumbrance, acceptable upside cap, and collateral process before quoting.

3.3 Miner production hedge

Client: Public/private Bitcoin miners.

Structure: Hedge a conservative slice of expected production or unencumbered BTC inventory using puts, put spreads, collars, or limited forwards.

Purpose: Protect cash-flow runway for power, capex, debt service, and ASIC/site investment.

Variants:

- monthly production collar;
- 3–6 month put spread on conservative production forecast;
- treasury BTC collar on 10–30% of unencumbered inventory;
- BTC-backed loan collateral protection;
- power/hashprice stress-window hedge.

Revenue model: Advisory/structuring, recurring execution, monitoring/reporting retainer.

Key control: Do not overhedge production. Size to conservative output, not optimistic hashrate targets.

3.4 Cross-venue BTC vol spread package

Client: Hedge funds, macro/RV funds, market makers, internal prop sleeve.

Structure: Relative-value packages across ETF options, CME, Deribit, and OTC.

Examples:

- ETF-vs-Deribit ATM vol spread;
- skew/risk-reversal spread;
- calendar spread;
- CME futures/options basis-vol package;
- event-vol package;
- client-flow hedge recycling.

Revenue model: Analytics subscription, execution/RFQ commission, later principal spread.

Key control: Only promote as an investor return engine after quote-verified or trade-verified evidence.

3.5 Structured notes / defined-outcome products

Client: Private banks, structured-product distributors, family offices, wealth platforms.

Structure: BTC-linked reverse convertibles, autocallables, buffered notes, principal-protected notes, income notes.

Purpose: Package BTC volatility into defined yield/upside/downside profiles.

Revenue model: Structuring margin, hedge spread, distribution economics.

Key control: Phase 2 only. Legal, suitability, issuer, disclosure, and valuation complexity is materially higher.

4. Data and analytics engine

4.1 Verified and available sources

Deribit: Verified public API. Live BTC options book summary returned ~936 contracts across 12 expiries in the first monitor pass. Fields include bid/ask/mid/mark, mark IV, underlying price, OI, volume, and instrument metadata.

IBIT/Nasdaq: Public Nasdaq JSON returned IBIT option chain records and bid/ask data. This is usable for pilot research but semi-official/undocumented. Production-grade use requires OPRA/vendor data.

BlackRock/iShares: Official holdings source supports BTC/share normalization. Previously verified official values showed BTC/share around 0.000564212717166, or roughly 1,772.38 IBIT shares per BTC, using observed BTC holdings and shares outstanding.

OCC: Official listed-series/OI source, useful for series validation but not quotes/IV.

CME: Official API/feed paths exist, but public pages should not be scraped. Production access requires credentials/licensing, DataMine, WebSocket/MDP feed, broker/vendor access, or terminal export.

OTC/RFQ: Not yet verified. Must be quote-verified before economics are treated as executable.

4.2 First quantified screen-only monitor

On the first v0 monitor, Deribit ATM IV term structure was roughly:

- DTE 1: ~31.3%
- DTE 4: ~30.1%
- DTE 8: ~34.9%
- DTE 15: ~36.3%
- DTE 43: ~38.0%
- DTE 134: ~40.8%
- DTE 225: ~43.7%

IBIT ATM IV estimates from Nasdaq mid prices and a rough Black-Scholes model were roughly:

- DTE 1: ~41.5%
- DTE 4: ~34.1%
- DTE 8: ~36.9%
- DTE 15: ~37.9%

Initial screen-only comparison:

- DTE 1: IBIT richer than Deribit by ~10 vol points.
- DTE 4: IBIT richer by ~4 vol points.

- DTE 8: IBIT richer by ~2 vol points.
- DTE 15: IBIT richer by ~1.5–2 vol points.

This does not prove arbitrage. It proves there are screen-level differences worth quote verification.

4.3 Monitor output standard

Every candidate must show:

- dislocation type: wrapper, skew, calendar, basis, margin/collateral, flow;
- exact legs;
- native prices;
- BTC-equivalent normalized prices;
- gross spread;
- estimated costs;
- net spread;
- size/capacity;
- client application;
- risk notes;
- source confidence;
- execution confidence.

Confidence labels:

- **Hypothesis:** theory only.
- **Screen-only:** visible from public/vendor screens.
- **Quote-verified:** indicative counterparty quote received.
- **Trade-verified:** executed trade or firm executable quote.

5. Natural client map

5.1 BTC treasury companies — first wedge

Examples: Strategy/MicroStrategy, Metaplanet, Twenty One Capital, Semler Scientific, KULR, The Blockchain Group/Capital B, BTC-heavy miners.

Pain: Capital raises, dilution optics, preferred/convert obligations, operating runway, BTC NAV volatility.

Product fit: Covered calls, zero-cost collars, put-spread collars, policy/reporting package.

Hook:

> Monetize a capped, board-approved slice of BTC convexity to fund obligations or reduce dilution, while preserving strategic BTC ownership and adding downside disaster protection.

5.2 Bitcoin miners — second wedge

Examples: MARA, Riot, CleanSpark, Cipher, TeraWulf, IREN, Core Scientific, Hut 8, Bitfarms, Bitdeer.

Pain: Power costs, capex, ASIC purchases, debt/refi needs, BTC-backed loans, production variability.

Product fit: Production collars, put spreads, inventory collars, runway protection.

Hook:

> Protect cash-flow runway for power, capex, and debt service by collaring a conservative slice of expected production or unencumbered BTC inventory, while leaving most upside intact.

5.3 RIAs, family offices, and wealth platforms

Pain: BTC drawdown tolerance, client income needs, custody restrictions, ETF workflow preference.

Product fit: ETF-option covered calls, collars, defined-loss call spreads, option-income SMA.

Hook:

> Add a volatility overlay to BTC ETF exposure without changing custody or introducing crypto exchange operations.

5.4 Hedge funds / macro / RV funds

Pain: Need normalized cross-venue data, execution proof, margin-adjusted edge.

Product fit: RV spread packages, analytics subscription, RFQ sourcing.

Hook:

> We normalize ETF, CME, Deribit, and OTC wrappers for basis, margin, collateral, bid/ask, and executable size, then label opportunities by quote/trade confidence.

5.5 Structured-product desks / private banks

Pain: Need defined BTC payoff products and hedge-source transparency.

Product fit: Autocallables, reverse convertibles, buffered notes, principal-protected notes.

Hook:

> Provide hedge-source transparency and BTC vol intelligence for defined-outcome BTC products.

6. Operating model

Phase 1: Research and analytics platform

- Build daily BTC Vol Desk Monitor.
- Ingest Deribit, IBIT, ETF holdings, OCC, and eventually CME/vendor data.
- Produce screen-only dislocation board.
- Create treasury/miner case studies.
- Maintain evidence appendix.

Revenue possibility: Research/analytics/advisory only, if legally structured.

Phase 2: Advisory and structuring

- Build board-ready treasury and miner hedging policies.
- Produce sample term sheets.
- Advise on permitted instruments, notional limits, collateral process, reporting, and risk governance.
- Partner with regulated execution counterparties.

Revenue possibility: Advisory retainer, structuring fee, consulting/project fee.

Phase 3: RFQ and execution workflow

- Source executable quotes from listed venues, brokers, OTC desks, or market makers.
- Compare quote-verified economics to screen economics.
- Provide best-execution-style analysis where legally permitted.

Revenue possibility: Commission, spread, referral/economic sharing where legally allowed.

Phase 4: Risk intermediation / principal sleeve

- Selectively warehouse/recycle risk across venues.
- Hedge client flow using ETF/CME/Deribit/OTC instruments.
- Operate inside proper fund, dealer, CTA/CPO, broker, or other legal wrapper as determined by counsel.

Revenue possibility: Principal spread, carry, management/performance fees.

Phase 5: Structured products

- Work with issuers/distributors to create BTC-linked defined-outcome products.
- Use analytics engine for hedge sourcing and pricing support.

Revenue possibility: Structuring margin, hedging spread, distribution economics.

7. Economics model

Potential revenue lines

1. ****Advisory/structuring fees****

- Treasury policy design.
 - Miner hedge design.
 - Board memo and risk framework.
2. ****Monitoring/reporting retainer****
- Daily/weekly hedge marks.
 - Exposure reporting.
 - Risk dashboard.
 - Board/lender reporting package.
3. ****Execution economics****
- Commission or spread through permitted broker/dealer/OTC relationships.
 - RFQ sourcing fee if legally allowed.
4. ****Principal/risk spread****
- Later-stage only.
 - Requires legal wrapper, capital, risk limits, and controls.
5. ****Analytics subscription****
- Hedge funds, allocators, internal desks.
 - Requires production-grade data and history.
6. ****Structured-product economics****
- Phase 2+.
 - Requires issuer/distribution/legal framework.

Unit economics to model next

For each sample structure:

- notional BTC or ETF share exposure;
- premium generated or paid;
- annualized yield;
- option bid/ask spread;
- hedge cost;
- RFQ spread;
- collateral/margin cost;
- reporting/advisory fee;
- maximum loss/upside cap;
- expected client value vs unhedged BTC;
- desk gross and net revenue.

8. Legal and regulatory perimeter

This is the main gating issue. The memo must not overclaim.

Potential activities touch multiple regimes:

- securities/options advice;
- commodity/futures/options advice;
- broker-dealer or introducing-broker activity;
- CTA/CPO analysis;
- investment adviser issues;
- OTC derivatives/ISDA counterparty activity;
- offshore exchange access;
- structured-product issuance/distribution;
- custody/collateral/prime brokerage arrangements.

Correct phased stance

Start with:

- research;
- analytics;
- hypothetical/screen-only monitor;
- education;
- internal memo;
- sample term sheets clearly marked illustrative;
- legal analysis.

Do not start with:

- quoting derivatives as principal to public companies;
- taking discretion over treasury assets;
- acting as broker-dealer/FCM/IB without structure;
- implying executable RFQ economics before quotes;
- offering securities/commodity advice without counsel.

Legal workstream deliverables

- regulatory activity map;
- permitted MVP activities;
- required disclaimers;
- entity structure options;
- partner/counterparty model;
- CTA/CPO/RIA/broker-dealer/IB analysis;

- OTC documentation checklist;
- public-company derivatives-policy template.

9. Risk controls

Market risk

- BTC delta and gamma.
- Volatility shifts.
- Skew/correlation.
- Calendar mismatch.
- Gap risk and weekend/24-7 risk.

Basis/wrapper risk

- ETF vs BTC NAV/premium-discount.
- CME futures basis.
- Deribit index/settlement differences.
- ETF options assignment and share settlement.

Liquidity risk

- Screen liquidity vs executable size.
- Wide short-dated markets.
- Stress-day spread widening.
- Position limits.

Margin/collateral risk

- USD vs BTC collateral.
- FCM margin calls.
- Crypto exchange margin/liquidation.
- OTC CSA terms.
- Public-company collateral restrictions.

Counterparty/venue risk

- Exchange/clearing/custody risk.
- OTC counterparty credit.
- Prime broker/FCM dependency.
- Offshore venue restrictions.

Operational/legal risk

- Inappropriate advice or execution without proper wrapper.
- Board/auditor/lender non-approval.
- Disclosure/regulatory issues.
- Valuation/model governance failure.

Narrative/reputation risk

- Treasury companies may view hedging as anti-BTC.
- Covered calls can be attacked if BTC rallies through strikes.
- Public companies need careful disclosure and policy framing.

10. 30-day pilot

Week 1 — data spine

Deliverables:

- Deribit BTC options ingestion.
- IBIT option-chain ingestion.
- iShares BTC/share parser hardened.
- OCC series/OI cross-check.
- CME data-source decision.
- Daily screen-only BTC Vol Desk Monitor v0.

Success criteria:

- Produce ATM IV term structure for Deribit and IBIT.
- Produce first normalized screen-only dislocation board.
- Every output labeled with source and execution confidence.

Week 2 — product prototypes

Deliverables:

- Treasury covered-call calculator.
- Zero-cost collar calculator.
- Put-spread collar calculator.
- Miner production hedge template.
- Sample term sheets.
- Board memo template.

Success criteria:

- One hypothetical BTC treasury case study.
- One hypothetical miner case study.
- All prices labeled illustrative/screen-only.

Week 3 — client mapping and commercial validation

Deliverables:

- 20–50 target accounts.
- Decision-maker/persona map.
- Outreach hooks.
- Qualification checklist.
- Counterparty/RFQ list.

Success criteria:

- At least 3 client archetype packages.
- At least 3 standard RFQ templates.
- Identify legal gating questions before any outreach.

Week 4 — investor memo and quote verification plan

Deliverables:

- Investor memo v1.
- Evidence appendix.
- Data architecture.
- Legal perimeter memo draft.
- RFQ verification plan.
- 90-day roadmap.

Success criteria:

- Clearly separate thesis, screen evidence, quote evidence, and execution requirements.
- No unsupported economics in investor-facing materials.

11. 90-day roadmap

Days 31–45

- Integrate CME via vendor/broker/terminal export or official credentialed path.
- Add OPRA-grade ETF option data if budget allows.
- Store daily surfaces historically.

- Add skew/risk-reversal and calendar analytics.
- Validate IBIT IV model against vendor or broker data.

Days 46–60

- Collect indicative RFQs for standard structures:
- 1-month covered call;
- 3-month zero-cost collar;
- 6-month miner put spread;
- ETF vs Deribit vol spread.
- Convert screen-only dislocation board into quote-verified board.
- Build first counterparty matrix.

Days 61–75

- Build treasury/miner reporting package.
- Draft legal operating model with counsel.
- Define entity/registration/partner structure.
- Create first client-ready board deck.

Days 76–90

- Run 3–5 live commercial conversations if legal perimeter permits.
- Finalize investor deck.
- Decide whether next vehicle is analytics/advisory company, fund, desk partnership, or hybrid.
- Define capital, hires, vendors, and counterparties required for launch.

12. Team and infrastructure needs

Team

- BTC/options structurer.
- Quant/data engineer.
- Derivatives legal counsel.
- Institutional sales / capital markets lead.
- Operations/risk person with FCM/prime/OTC experience.

Infrastructure

- Market data: Deribit, OPRA/vendor, CME/vendor.
- Data store for option surfaces and historical marks.
- Pricing library and validation process.

- RFQ/counterparty tracker.
- Risk dashboard.
- Compliance archive.
- Board/client report generator.

Counterparties

- FCM / CME access.
- OPRA/options data vendor.
- ETF/options broker.
- OTC crypto options desks.
- Custodian/prime broker.
- Legal/accounting advisers.

13. Investment case

What makes the idea attractive

- BTC volatility is structurally high and institutionally relevant.
- BTC exposure now exists across fragmented wrappers.
- Public BTC treasuries and miners create natural hedging/yield flows.
- ETF/CME/Deribit/OTC access constraints create intermediation value.
- The first product is understandable: treasury income and protection.
- The analytics engine can become a proprietary pricing and risk-recycling layer.

What must be proven

- Screen dislocations are executable after costs.
- Clients will accept capped upside on a defined tranche.
- Legal structure permits advisory/execution economics.
- Data pipeline can produce reliable normalized surfaces.
- Counterparties will quote size.
- Risk can be hedged across wrappers under stress.

Main red flags

- Legal/regulatory perimeter may be heavier than expected.
- Pure vol-arb edge may be competed away.
- Public companies may avoid derivatives due to optics.
- OTC/RFQ spreads may consume apparent screen edge.
- Margin/collateral costs may erase relative-value profits.

- Deribit access may be unusable for key US clients.

14. Recommended investor-facing positioning

Lead with:

> We help institutional BTC holders transform volatility into income, protection, and structured risk through board-ready treasury and miner hedging programs, powered by a proprietary cross-venue BTC volatility monitor.

Avoid leading with:

> We run a BTC options arbitrage fund.

The former is a business. The latter is a trade.

15. Immediate next deliverables

1. Harden the data pipeline:

- fix iShares BTC/share parser;
- store Deribit/IBIT snapshots;
- compute IV consistently;
- add bid/ask filters.

2. Build first treasury case study:

- 10,000 BTC hypothetical holder;
- 10% covered-call sleeve;
- 3-month collar;
- 6-month put-spread collar;
- board-policy summary.

3. Build first miner case study:

- 3-month conservative production forecast;
- 25–50% hedge ratio;
- put-spread/collar;
- cash runway impact.

4. Start legal perimeter memo:

- what can be done as research;
- what requires registration/partner;
- what cannot be done before counsel.

5. Start RFQ verification plan:

- list standard structures;
- identify counterparties;
- define quote fields;
- label quote-verified vs screen-only.

Bottom line

Idea 1 is viable enough to continue, but only if built in the disciplined order:

1. evidence spine;
2. screen-only monitor;
3. treasury/miner product prototypes;
4. legal perimeter;
5. quote verification;
6. investor memo;
7. execution vehicle.

The strongest launch product is not a prop-vol fund. It is a board-ready BTC treasury and miner hedging platform with a proprietary cross-venue volatility engine behind it.